

# Part Two: Factor Analysis and Structural Equation Modeling

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## 1 What Part Two Is

Part Two of *Quantitative Methods for Social and Behavioral Research with R* introduces latent-variable modeling through factor analysis and prepares the transition to structural equation modeling. It is written for postgraduate students and applied researchers who need to understand what a latent-variable model asserts, how its parameters are estimated, and how far its results support a substantive interpretation.

The first four chapters form one measurement-model sequence. Chapter 1 introduces the common-factor model and its representations. Chapter 2 develops an exploratory factor model. Chapter 3 turns the resulting proposal into a restricted confirmatory model and estimates it. Chapter 4 evaluates that fit, compares planned rivals, studies a revision, and introduces advanced measurement structures. The sequence is cumulative: later chapters assume the model, notation, and analytic history established earlier.

This is not a catalogue of software commands or fit-index cutoffs. Equations, matrices, diagrams, prose, and code are treated as different representations of one model. Software makes the calculations reproducible; it does not choose the research question, justify a construct label, or turn an exploratory decision into an independently confirmed result.

**What is assumed.** Variance, covariance, correlation, sampling uncertainty, and elementary regression notation at the level developed in Part One. Basic matrix multiplication helps, but the operations each model needs are explained where they are used. Calculus and proof-based mathematical statistics are not prerequisites.

## 2 The Measurement Problem at a Glance

Several observed indicators may vary together even though the attribute used to explain their association is not recorded directly. A common-factor model represents that proposal as

$$\begin{aligned} \mathbf{y}_i &= \nu + \Lambda\eta_i + \epsilon_i, & E(\mathbf{y}_i) &= \nu + \Lambda\alpha, \\ & & \Sigma(\vartheta) &= \Lambda\Phi\Lambda^T + \Theta. \end{aligned}$$

Here  $\mathbf{y}_i$  contains observed indicators,  $\eta_i$  unobserved factors,  $\Lambda$  loadings, and  $\epsilon_i$  what the declared factors leave unexplained;  $\nu$  and  $\alpha$  are indicator and latent intercepts, and  $\vartheta$  collects the free parameters. The covariance equation divides observed variation into a common part and a residual part, and gives the model observable consequences: proposed loadings, factor relations, and residual restrictions imply a particular covariance matrix.

The equation does not make a factor identical to the construct used to name it. A factor is an unobserved random variable inside a statistical model; a construct is a substantive concept whose interpretation also depends on the indicator domain, response process, population, design, and intended use. Likewise, a factor-to-indicator arrow states the direction of representation, not that the factor causally produces a response. Good covariance fit can support a statistical representation without showing that a latent entity exists as named.

Part One's question–answer discipline stays in the background: a method should answer a declared question, and the conclusion should not outrun its evidence. Chapter 1 contains one optional application of that vocabulary to the latent-variable case; Chapters 2–4 concentrate on the statistical work.

## 3 How Chapters 1–4 Progress

The four chapters make four connected moves: **represent** a common-factor model and derive its implied moments; **explore** a development record to settle dimensionality, orientation and interpretation; **specify and estimate** the resulting proposal as explicit restrictions on unused data; and **evaluate, compare and extend** the fitted model. The covariance equation is the same throughout; what changes is the responsibility assigned to it.

Figure 1 makes the architecture explicit. Each stage changes the question and evidence role while retaining the same common-factor foundation.

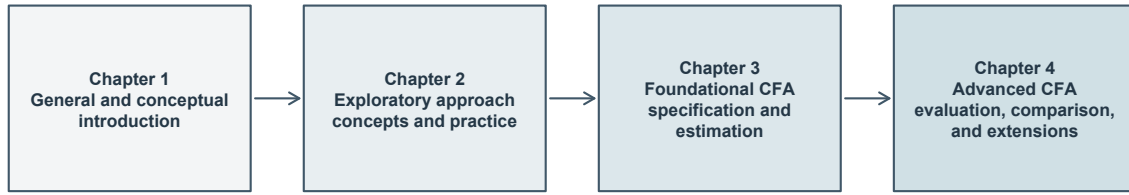


Figure 1: The four-chapter architecture moves from representation to exploratory development, foundational CFA, and advanced CFA.

Table 1 summarizes the responsibility and handoff of each chapter.

**Table 1**

The four-chapter measurement-model sequence.

| Chapter                                                     | Main responsibility                                                                                                                                                                                                                                             | What it hands forward                                                                                                                      |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. From Latent-Variable Modeling to Factor Analysis</b>  | Distinguish observed variables, factors, constructs, composites, and scores; translate the common-factor model among scalar equations, matrices, diagrams, and prose; derive implied means and covariances; explain residuals and latent-coordinate conventions | A technically explicit common-factor representation, known-value teaching records, and the questions that exploratory analysis must answer |
| <b>2. Exploratory Factor Analysis and Model Development</b> | Audit indicators and their correlations; choose an extraction target; investigate factor count; rotate and interpret neighboring solutions; examine residuals, sensitivity, and stability                                                                       | A documented three-factor candidate, named alternatives, and unresolved questions frozen before unused confirmation data are inspected     |

| Chapter                                                        | Main responsibility                                                                                                                                                                                                                                                                                                                              | What it hands forward                                                                                                   |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>3. Specifying and Estimating Confirmatory Factor Models</b> | Translate the candidate into fixed, free, and constrained parameters; set factor metrics; count moments and coordinates; examine identification with small-model diagrams; translate a published four-factor model into syntax and a parameter ledger; estimate by continuous ML; read convergence, admissibility, coefficients, and uncertainty | A reproducible fitted object and parameter ledger whose overall adequacy has deliberately not yet been decided          |
| <b>4. Evaluating and Comparing Confirmatory Factor Models</b>  | Read global fit as a profile; locate local discrepancy; compare compatible planned rivals; interpret modification diagnostics; freeze and revalidate a revision; fit and compare higher-order and bifactor representations; conduct a retrospective known-population audit; examine reliability and score boundaries                             | A calibrated measurement-model conclusion and a bridge from relations among factors to later structural equation models |

The stopping points are deliberate. Chapter 2 does not relabel a developed pattern as confirmation; Chapter 3 does not treat convergence as adequate fit; Chapter 4 does not treat favorable fit as construct validation. Each chapter completes one responsibility before transferring a defined record to the next.

Four mathematical threads run through the sequence:

- **Common and residual variation.** Scalar covariance expansions become the matrix decomposition  $\Lambda\Phi\Lambda^\top + \Theta$ , then fitted residuals.
- **Coordinates and restrictions.** Scale, sign and rotational freedom lead to rotation, scaling constraints, identification, and the distinction between equivalent and testably different models.
- **Information and estimation.** Distinct moments, free coordinates and candidate degrees of freedom are counted consistently before any numerical output is read.
- **Evaluation and use.** Parameter uncertainty, global fit, local residuals, planned comparisons, reliability and scores answer different questions, and none is a pass–fail certificate for a model or a construct.

Chapter 1 needs no student software. Chapter 2 introduces the `psych` and `GPArotation` workflow; Chapters 3–4 use `lavaan`. In every case the printed chapter supplies enough to follow the model and the conclusion without running the code.

## 4 Evidence Roles Across the Sequence

Repeated use of one topic does not mean that every record has the same inferential role. Part Two makes the analysis history visible so that development, first evaluation, and revalidation are not collapsed into one favorable story. Table 2 gives the principal roles.

**Table 2**

Records keep distinct analytical roles even when they illustrate the same model family.

| Record                                   | First permitted use                                                                 | What it cannot establish by itself                                                                     |
|------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Known-value six-indicator records</b> | Chapter 1 reconstructs covariance from declared parameters                          | Sampling behavior, empirical factor count, or the existence of named constructs                        |
| <b>SLE Sample A, 360 cases</b>           | Chapter 2 develops and freezes the exploratory candidate                            | Independent confirmation of choices made while examining A                                             |
| <b>SLE Sample B, 360 cases</b>           | Chapter 3 estimates the frozen candidate; Chapter 4 evaluates it and planned rivals | Independent support for a new relation first suggested by B                                            |
| <b>SLE Sample C, 360 cases</b>           | Chapter 4 evaluates one revision frozen after B                                     | General validity, invariance, or permission to search repeatedly in C                                  |
| <b>Known SLE generating moments</b>      | Chapter 4 reveals them retrospectively, after the B/C analysis history is complete  | Contemporaneous justification for earlier choices, or finite-sample bias inferred from one realization |
| <b>Separate donor microcases</b>         | Compact demonstrations of specified calculations and model comparisons              | The development or confirmation status of the SLE sequence                                             |

The Student Learning Experiences (SLE) records are synthetic continuous data, not observations from actual students: three independent samples of 360 cases on twelve indicators – four Planning, four Strategy and four Belonging. Each chapter simulates the sample it is entitled to from a declared population and a fixed seed, and the Chapter 2 appendix supplies the codebook and manifest. The samples are never pooled to manufacture a larger analysis, and a record that becomes exploratory through revision does not recover confirmatory status because the next command is called CFA.

The separation is a teaching device as much as a discipline. It lets readers see that a model can be coherent, reproducible and apparently well fitting while still owing a narrower conclusion than its name suggests.

## 5 Boundaries and What Follows

Part Two treats indicators as continuous and keeps one coherent continuous-ML lineage. Ordered categorical indicators, thresholds, polychoric correlations and their estimator-specific conventions are signposted but taught elsewhere. The boundary does not exclude categorical grouping variables or coded predictors, which later multigroup and structural models may use while retaining continuous indicators.

The architecture covers reflective common-factor measurement models. It does not make a factor score an error-free observation, equate fit with validity, or establish causality from a path diagram; higher-order and bifactor models appear as rival representations, not as remedies for a weak core model. Item-response theory, mixture models, multilevel factor analysis and a full treatment of categorical indicators belong elsewhere.

After Chapter 4, Part Two turns from relations between factors and indicators to relations among variables in a structural system: path analysis, general structural equation modeling, measurement invariance, and latent growth. Multilevel growth belongs to Part Three. The notation is chosen to survive that transition – a latent intercept is not generally the marginal latent mean once structural regressions enter – so later chapters extend the model rather than quietly changing what its symbols mean.

Each numbered chapter closes with a **Summary and reading bridge**, followed by unnumbered exercises. Appendices may add derivations or software details, but the principal question and its calibrated answer remain understandable from the numbered route.